

LLM-Guided ANN Index Optimization for Efficient Human-Object Interaction Retrieval

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Abstract

Retrieval systems underpin modern AI applications — spanning visual search, recommendation engines, and multimodal question answering at billion-scale. Modern multi-stage retrieval systems require the joint optimization of highly coupled parameters, yet traditional hyperparameter optimization (HPO) methods — including Bayesian TPE and Gaussian Process BO — rely on an independence assumption that fundamentally prevents them from navigating these entangled configuration spaces. We address this limitation with a phase-aware LLM agent that operates as a feedback loop — learning from its own optimization history to navigate the coupled parameter space across self-regulated exploration, exploitation, and fine-tuning phases. Evaluated on the HICO-DET human-object interaction retrieval benchmark using Intel VDMS (Visual Data Management System), our agent outperforms Optuna TPE by +33.3% and GP-BO by +50.7% in accuracy-throughput score, delivering a 15.3× throughput gain over UniIR. Validation across three benchmarks confirms that the agent’s advantage grows with the degree of parameter coupling: +33.3% on HICO-DET (high coupling), +1.2% on GLDV2 (moderate coupling), and < +0.1% on SIFT1M (near-independent control). These results position our agent as the method of choice for retrieval systems where parameter coupling is the dominant source of optimization complexity.

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1 Introduction

The throughput and accuracy of a vector database management system (VDBMS) depend critically on ANN index configuration: on a production-scale deployment, the gap between default and optimized settings can exceed an *order of magnitude* in queries per second on identical hardware. Such systems underpin modern AI applications — product recommendation, visual question answering, medical image retrieval, and cross-modal content discovery — each issuing millions of queries per day. Closing this gap demands an optimizer that reasons over how parameters interact across retrieval stages, not one that searches blindly.

Multi-stage retrieval pipelines compound this challenge: systems such as VDMS [27], Milvus [28], and GaussDB-Vector [25] chain an ANN retrieval stage with a cross-modal fusion reranking stage, exposing interdependent index and reranking parameters whose joint configuration determines both retrieval quality and throughput. These parameters are structurally coupled: reducing retrieval search depth (*efSearch*) forces the fusion reranker to compensate via a higher fusion weight (α) — a cross-stage dependency invisible to any parameter-independent optimizer. On HICO-DET, the gap between an unoptimized baseline and our agent-optimized configuration reaches 36.6× in throughput on identical hardware — on a single A40 GPU node, the difference between 8 and 300 queries per second, between a research prototype and a production-ready retrieval service.

Yet no existing method is designed to close it. On the *LLM-guided configuration* side, GPTuner [12] and λ -Tune [7] demonstrate that LLM-guided methods outperform ML-based [26] and reinforcement-learning baselines on DBMS knob tuning. On the *ANN index tuning* side, VDTuner [32] applies multi-objective Bayesian optimization to VDMS index parameters but treats them independently, as do standard search baselines — Optuna TPE [1], grid, and random search [3, 23].

Yet all share a structural limitation: LLM-guided DBMS tuners operate on flat knob spaces (PostgreSQL, MySQL) where parameters are independent — no cross-stage coupling exists by construction. The ANN tuning methods — Optuna TPE [1], grid and random search [3, 23] — treat each parameter independently, missing this

cross-stage coupling; VDTuner [32], the closest prior system, adds a further failure mode when its smooth surrogate model encounters the hard feasibility threshold our metric imposes.

We propose a *phase-aware LLM agent* for coupled ANN index optimization, applicable to any VDBMS exposing tunable index parameters and instantiated in this work on Intel’s VDMS [27]. The agent conditions each proposal on all prior configurations and their observed scores, reasons explicitly over cross-stage parameter dependencies, and self-regulates between exploration and exploitation phases. Text-query HOI retrieval — returning images of specific human-object interactions in response to natural language queries — serves direct applications in video surveillance, assistive robotics, and interactive multimodal search. We also introduce the *first text-query image retrieval benchmark* from HICO-DET [5] — 47,776 images, 600 HOI categories, deployed on VDMS — the first benchmark to expose the structural limits of blind ANN optimizers, as every prior use of HICO-DET is restricted to the detection paradigm (e.g., [11, 14, 19, 33]).

Evaluated under the SIEVE quality-constrained throughput objective (Section 6.1), the agent achieves +33.3% over Optuna TPE and +50.7% over GP-BO on HICO-DET, delivering a 36.6× improvement over an exhaustive-search CLIP baseline [9]. On simpler datasets — GLDv2 [30] (image-to-image) and SIFT1M [8] (pure ANN search) — the advantage diminishes proportionally, confirming it scales with structural configuration coupling.

Contributions. This paper makes the following contributions:

- **Phase-aware LLM agent for coupled ANN index optimization.** A history-conditioned agent applicable to any VDBMS exposing ANN index parameters (instantiated on Intel’s VDMS) that explicitly models the $efSearch-\alpha$ inter-stage coupling, discovering compensatory operating points that no parameter-independent method locates within 50 iterations. Achieves +33.3% over Optuna TPE and +50.7% over GP-BO on HICO-DET; mechanistic ablations confirm history conditioning and phase-aware transitions are each independently necessary.
- **First HOI retrieval benchmark and zero-indexing-cost two-stage pipeline.** The first text-query image retrieval benchmark from HICO-DET (47,776 images, 600 verb-object categories, deployed on VDMS), pairing CLIP ANN retrieval with DINOv2 pseudo-relevance-feedback reranking at zero additional indexing cost (+3.65 pp mAP at <1% extra latency). This demonstrates that visual diversity features complement semantic alignment without additional indexing or storage overhead, enabling quality improvement at effectively zero deployment cost.
- **Complexity-matched evaluation of when LLM reasoning pays off.** We characterize when LLM-guided optimization pays off across three datasets spanning the full coupling-complexity spectrum: fully coupled 7-parameter cross-stage (HICO-DET), image-to-image (GLDv2), and pure ANN search (SIFT1M) — showing that gains diminish proportionally as structural coupling decreases. This gives practitioners a principled criterion for when LLM-guided optimization justifies its overhead over simpler baselines.

- **SIEVE: a quality-constrained throughput objective for deployment-aware ANN evaluation.** We propose SIEVE, a quality-constrained throughput metric that scores zero for any configuration below a minimum retrieval quality threshold, reflecting real deployment requirements; we show empirically that product-metric-optimal configurations systematically fall below this threshold across the majority of methods and seeds — demonstrating that smooth metrics steer optimization into the infeasible region.

The rest of this paper is organized as follows: Section 2 surveys related work; Sections 3 and 4 introduce the HICO-DET retrieval benchmark and two-stage pipeline; Section 5 details the LLM-guided optimizer; Section 6 reports experiments; and Section 7 concludes.

2 Related Work

2.1 Vision-Language Retrieval and HOI Detection

HOI detection has produced a series of strong transformer-based systems since HICO-DET [5] established the 600-category verb-object benchmark. HOTR [11] introduced end-to-end interaction detection via encoder-decoder architectures, and CDN [33] demonstrated complementary gains from combining one-stage and two-stage detection. HOICLIP [19] showed that vision-language pre-training transfers interaction knowledge efficiently, while DPHOI [14] exploited disentangled pre-training over interaction primitives. All operate in the detection paradigm — predicting (person, object, interaction) triples from an image — and the sole prior non-detection use of HICO-DET [10] employs spatial canvas composition with IoU matching rather than free-form text queries.

CLIP [21] established joint vision-language embedding via contrastive pre-training on 400 million image-text pairs, enabling zero-shot text-to-image retrieval. DINOv2 [20] produces self-supervised visual features that complement CLIP’s language-aligned representations, capturing fine-grained visual structure suited to reranking. UniIR [29] unified eight multimodal retrieval tasks under a single model trained on M-BEIR, establishing a strong cross-modal baseline. LamRA [16] demonstrated that large multimodal models further improve fine-grained query understanding for retrieval. We fill the orthogonal HOI *retrieval* direction and adopt CLIP and UniIR as primary baselines.

2.2 Vector Database Systems and ANN Indexing

The efficiency of HOI retrieval depends not only on the query models above but also on how the underlying approximate nearest-neighbor indices are configured. ANN indexing algorithms span graph-based indices (HNSW [17] via hierarchical navigable small-world graphs, DiskANN [24] for billion-point SSD-resident datasets), inverted file structures with product quantization [8], and GPU-accelerated flat search [9]. ANN-Benchmarks [2] standardizes recall-versus-throughput evaluation across these algorithms; the NeurIPS’21 competition [22] established billion-scale baselines.

Production VDBMSs integrate ANN indexing with metadata management and query processing. Milvus [28] supports multiple index backends with a cloud-native query layer. GaussDB-Vector [25] decouples index construction from query serving to sustain low latency at billion-vector scale. VSAG [35] enables incremental updates via subgraph monotonicity, eliminating costly full rebuilds. VDMS [27] co-locates graph-property metadata with vector data and exposes HNSW index parameters through a Python API, making it the natural platform for an optimizer that must rebuild and re-query the index at each evaluation step. Across all these systems, index configuration is left to manual tuning or blind search; automated optimization for multi-stage coupled parameter spaces remains an open problem.

2.3 Hyperparameter Optimization

Standard HPO methods include random search [3], which provides a strong baseline under bounded evaluation budgets; Bayesian optimization with Gaussian Process surrogates [23], which applies acquisition functions to balance exploration and exploitation; and bandit-based scheduling such as Hyperband [13], which allocates evaluation budget adaptively. Auto-sklearn [6] combines Bayesian optimization with meta-learning over prior tasks to warm-start configuration search. Optuna [1] implements Tree-structured Parzen Estimators for mixed-type spaces and serves as our strongest non-LLM baseline.

For DBMS tuning, OtterTune [26] pioneered ML-guided knob selection via Gaussian Process regression over workload representations. VDTuner [32] extends this approach to multi-objective Bayesian optimization for VDMS HNSW parameters, improving over manual baselines, but models each parameter independently. All of these methods — standard HPO [1, 3, 13, 23], AutoML [6], and system-specific Bayesian optimization [26, 32] — treat parameters as mutually independent, a structural limitation that prevents them from discovering compensatory configurations across coupled retrieval stages.

2.4 LLM-Guided System Configuration

Recent work has explored whether large language models can serve as optimization agents, bringing natural language reasoning to configuration search. OPRO [31] established that LLMs can act as black-box optimizers by conditioning successive proposals on a history of candidate solutions and their scores. Follow-up work extended this paradigm to ML hyperparameter optimization [15, 34].

GPTuner [12] applies LLM-guided Bayesian optimization to PostgreSQL and MySQL knob tuning, extracting structured configuration knowledge from database documentation to guide search. λ -Tune [7] demonstrates cross-system generalization: LLMs can transfer configuration knowledge across DBMS platforms without system-specific retraining. Both outperform ML-based and RL-based baselines on DBMS tuning, yet both operate on flat, independently interpretable knob spaces where no cross-stage coupling exists by construction.

Our work addresses what neither thread has reached: a coupled multi-stage parameter space where ANN retrieval depth and fusion reranking weight must be optimized jointly under a hard feasibility constraint that smooth surrogate models cannot represent. A history-conditioned, phase-aware LLM agent is uniquely suited to this setting.

3 The HICO-DET Retrieval Benchmark

3.1 Source Dataset

HICO-DET [5] contains 47,776 images (38,118 train / 9,658 test) annotated across 600 HOI categories with per-instance bounding boxes and interaction labels, where each category is a *verb-object* pair drawn from 117 action verbs and 80 MS-COCO object classes. Three structural properties make HICO-DET particularly challenging for ANN index configuration optimization. First, *compositional fine-grained discrimination*: the 600 categories span 80 COCO object classes and 117 verb types; categories sharing the same object (e.g., *ride horse*, *sit_on horse*, *straddle horse*) differ only in the action verb, so single-encoder retrieval conflates them and the joint selection of ANN engine and Stage-2 fusion weight becomes critical. Second, *long-tail relevance*: category sizes are highly skewed, with the largest exceeding 1,000 positive images and the smallest having as few as 5; an optimizer that maximizes mean mAP without attending to per-category variance will exploit high-frequency categories while failing on rare but equally weighted queries. Third, a *dual-encoder modality gap* requires joint tuning: CLIP retrieves by language-aligned semantic similarity while DINOv2 captures fine-grained visual structure complementary to CLIP’s representations [20]; CLIP alone (Stage 1 without DINOv2 reranking) achieves lower mAP than the two-stage fusion, so the Stage-2 fusion weight α is a data-dependent parameter that must be tuned jointly with the Stage-1 ANN configuration.

3.2 Benchmark Construction

Retrieval Corpus. We use all 47,776 images as a single retrieval corpus. The original HICO-DET train/test split was designed for supervised detection training and carries no meaning for retrieval evaluation, where the task is to rank corpus images given a query, not to generalize across a detection train/test split. Merging both splits maximizes the positive-pair density needed for meaningful mAP computation across 600 categories; restricting evaluation to the 9,658-image test split alone would reduce per-category positive counts below the minimum threshold for reliable mAP computation across all 600 classes. CLIP ViT-L/14 [21] embeddings are pre-computed offline using the official OpenAI weights and stored in VDMS as a 768-dimensional L2 DescriptorSet of 47,776 vectors.

Query Design: One Canonical Query per HOI Class. We construct exactly one canonical natural-language text query per HOI category, yielding 600 queries. A single canonical query per category eliminates inter-query variance from paraphrase choice and ensures that each of the 600 HICO-DET categories contributes equally to the mean Average Precision score [18]. Each query follows the template “a person {verb} a {object}”, with the special case “a person near a {object}” for the no_interaction verb class. Queries are encoded with the same CLIP ViT-L/14 text

encoder, producing 768-dimensional unit-norm text embeddings that reside in the same embedding space as the indexed image vectors.

Ground Truth: Multi-Label Positive Structure. An image is relevant to a query if it carries the corresponding HOI annotation in the HICO-DET ground truth. A single image may be relevant to multiple queries (e.g., an image annotated with both *ride motorcycle* and *hold phone* contributes positives to two queries). Summed over all 600 queries, there are 90,641 positive (image, query) pairs. For each HOI category, the single image used as the DINOv2 reranking reference is excluded from relevance judgments (treated as a junk image), ensuring the reference cannot trivially self-retrieve; after this exclusion the minimum per-category positive count is 5. Every query has at least 5 relevant images, so mAP is well-defined for all 600 queries.

3.3 Evaluation Protocol

We evaluate retrieval quality using mean Average Precision (mAP), Precision@10, NDCG@10, and Recall@10 over all 600 queries. mAP is our primary metric: it rewards systems that rank relevant images higher in the retrieved list and is the standard metric for ranked retrieval with multiple relevant documents per query [18]. Throughput is measured as batch QPS: 600 queries issued as a single batch to a live VDMS container, divided by total wall-clock time. The joint optimization objective is

$$\text{Score}(\theta) = \begin{cases} \text{QPS}(\theta) & \text{if } \text{mAP}(\theta) \geq \tau, \\ 0 & \text{otherwise,} \end{cases} \quad \tau = 0.15. \quad (1)$$

This threshold-gated objective rewards throughput only within the feasible region ($\text{mAP} \geq \tau$): a configuration that sacrifices retrieval quality below τ scores zero regardless of how high its QPS is, while within the feasible region higher QPS is strictly better.

3.4 Problem Statement

Definition 1 (Configuration Space). Let Θ denote the joint configuration space of the two-stage pipeline (Section 4). Θ is the Cartesian product of seven parameter domains (*engine*; M , $nlist$; $efSearch$, $nprobe$; k ; α ; n_{refs} ; $ref_strategy$) enumerated in Table 1; $|\Theta| = 102,144$.¹

Definition 2 (Evaluation Oracle). For any configuration $\theta \in \Theta$, the evaluation oracle $O(\theta) = (\text{mAP}(\theta), \text{QPS}(\theta))$ is obtained by (i) rebuilding the VDMS ANN index under θ , and (ii) issuing all 600 HICO-DET text queries to the live container and recording mAP (Section 3) and batch QPS. The oracle is *black-box*: no closed-form gradient with respect to θ is available, and each call requires a full index rebuild followed by 600 retrieval operations.

Definition 3 (Optimization Problem). Given Θ (Definition 1), O (Definition 2), and a finite evaluation budget $N \in \mathbb{N}$, find

$$\theta^* = \arg \max_{\theta \in \Theta} \text{Score}(\theta), \quad (2)$$

¹The LLM agent additionally proposes a VDMS metadata pre-filter (*constraint_strategy* $\in \{\text{none}, \text{object}, \text{verb}, \text{object_and_verb}\}$) that restricts ANN search to images belonging to a specified COCO object or verb class before retrieval; this parameter lies outside Θ as defined here. Grid Search and Random Search fix *constraint_strategy* = none; Optuna TPE samples it as an additional categorical hyperparameter; the LLM agent uses *constraint_strategy* = object in all reported best configurations.

Table 1: Configuration space Θ (Definition 1); [†] conditional on engine; total $|\Theta| = 102,144$.

Parameter	Notation	Value Set	Card.
<i>Stage 1 — ANN retrieval</i>			
engine	—	Flat, HNSWFlat, IVFFlat	3
conn. depth [†] (HNSW)	M	8, 16, 32, 48, 64	5
cluster count [†] (IVF)	$nlist$	64, 128, 256, 512	4
search beam [†] (HNSW)	$efSearch$	32–500 (7 vals)	7
probed cells [†] (IVF)	$nprobe$	4, 8, 16, 32, 64	5
candidate pool	k	50–1000 (8 vals)	8
<i>Stage 2 — DINOv2 reranking</i>			
fusion weight	α	0.0, 0.05, ..., 0.90	19
ref. count	n_{refs}	1, 3, 5, 10	4
ref. strategy	—	first, centroid, diverse	3
Total Θ			102,144

subject to at most N calls to O , where $\text{Score}(\theta)$ is defined by Eq. (1) ($\text{QPS}(\theta)$ if $\text{mAP}(\theta) \geq \tau$, else 0). We set $N=50$ throughout, covering 0.05% of $|\Theta|$.

4 Two-Stage Retrieval Pipeline

4.1 Oracle Instantiation and Notation

Oracle O (Definition 2) maps any $\theta \in \Theta$ to a (mAP, QPS) pair; the optimizer maximizes $\text{Score}(\theta)$ (Eq. (1)): $\text{QPS}(\theta)$ if $\text{mAP}(\theta) \geq \tau$, else 0. Let $C = \{(c_i, d_i)\}_{i=1}^N$ denote the corpus of $N=47,776$ images, where $c_i \in \mathbb{R}^{768}$ is the unit-normalized CLIP ViT-L/14 embedding and $d_i \in \mathbb{R}^{1024}$ is the DINOv2 ViT-L/14-reg4 embedding of image i , and $\mathcal{I}_h \subseteq [N]$ the annotated relevant set for HOI class h . Given a text query q_h (e.g., “a person riding a horse”), the pipeline proceeds in two stages as illustrated in Figure 1.

Stage 1 encodes q_h into a unit-norm vector $q_c \in \mathbb{R}^{768}$ via the frozen CLIP ViT-L/14 text encoder and issues an ANN search over the VDMS CLIP DescriptorSet to retrieve a candidate pool $\mathcal{T}_k(q_h)$ of cardinality k . **Stage 2** reranks $\mathcal{T}_k(q_h)$ using DINOv2 visual distances computed against a pre-built per-class reference r_h , fusing both distance signals without any additional index. The seven parameters of θ — *engine*, M or $nlist$, $efSearch$ or $nprobe$, k , α , n_{refs} , $ref_strategy$ — govern the accuracy–throughput trade-off across both stages and constitute the search space Θ (Table 1). The oracle cost per call is dominated by index construction (≈ 425 – 455 s) and 600 batch queries (≈ 35 – 99 s), yielding ≈ 8 – 9 minutes of wall-clock time per oracle evaluation.

4.2 Stage 1: ANN Candidate Generation

All N CLIP ViT-L/14 image embeddings are pre-computed offline and stored in VDMS as a 768-dimensional L2 DescriptorSet. At query time, q_h is encoded by the frozen text encoder and submitted to VDMS as a FindDescriptor request. VDMS returns the top- k candidate identifiers together with their L2 distances $\delta_j^c = \|c_j - q_c\|_2$; the candidate pool is formally

$$\mathcal{T}_k(q_h) = \{j \in [N] : \|c_j - q_c\|_2 \leq \delta^{(k)}\}, \quad (3)$$

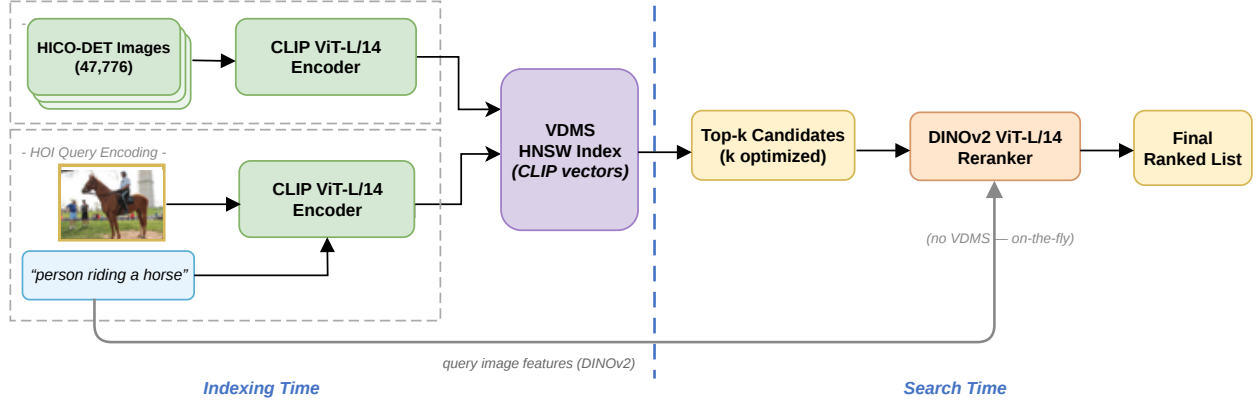


Figure 1: Two-stage retrieval pipeline. CLIP ViT-L/14 ANN retrieval (Stage 1) produces a top- k candidate pool; DINOv2 ViT-L/14-reg4 reranking (Stage 2) fuses scores via $s_j = (1-\alpha)\hat{c}_j + \alpha\hat{d}_j$ at zero additional indexing cost.

where $\delta^{(k)}$ denotes the k -th smallest L2 distance from q_c to any corpus embedding. For FaissFlat, Eq. (3) is satisfied exactly; for FaissHNSWFlat and FaissIVFFlat, VDMS returns a high-recall approximation whose tightness is governed by the engine-specific search parameters.

Three ANN engines expose qualitatively distinct accuracy-throughput operating points.

FaissFlat: exact exhaustive L2 search; recall upper bound at lowest QPS.

FaissHNSWFlat [17]: hierarchical navigable small-world graph; recall/latency controlled by graph connectivity M and beam width $efSearch$ (values in Table 1). The build-quality parameter $efConstruction$ is fixed at 200 throughout HICO-DET and GLDv2 experiments; preliminary experiments confirmed that values above 200 yield negligible recall improvement at substantially higher build cost, so it is excluded from Θ .

FaissIVFFlat [9]: inverted-file index; search probes $nprobe$ of $nlist$ Voronoi cells (values in Table 1).

The candidate pool size k (8 values: 50–1000, Table 1) determines the recall ceiling available to Stage 2 and is selected jointly with engine parameters. Batch QPS is defined as

$$QPS(\theta) = \frac{|Q|}{T_{total}(\theta)}, \quad (4)$$

where $|Q|=600$ is the number of HOI text queries and $T_{total}(\theta)$ is the elapsed wall-clock time for all queries issued in a single batch to the live VDMS container under configuration θ . This end-to-end latency encompasses the VDMS round-trip for Stage 1, the in-memory reranking computation in Stage 2, and all inter-process communication overhead.

4.3 Stage 2: Feature Fusion and Reranking

Stage 2 reranks $\mathcal{T}_k(q_h)$ using DINOv2 ViT-L/14-reg4 [20] visual features — capturing fine-grained appearance structure complementary to CLIP’s language-aligned semantics — without constructing

any additional index. Each HOI class h is represented by a visual reference vector $r_h \in \mathbb{R}^{1024}$, built offline as the mean DINOv2 embedding of a selected subset $S_h \subseteq \mathcal{I}_h$:

$$r_h = \frac{1}{|S_h|} \sum_{i \in S_h} d_i, \quad (5)$$

where $n_r = |S_h| \in \{1, 3, 5, 10\}$. For $n_r=1$ (the default), r_h is the DINOv2 embedding of the first image in the HOI category. For $n_r>1$, three strategies populate S_h : **first** takes the initial n_r images in dataset order; **centroid** selects the n_r images nearest to the DINOv2 class mean $\mu_h^d = \frac{1}{|\mathcal{I}_h|} \sum_{i \in \mathcal{I}_h} d_i$, anchoring r_h at the visual core of the class; **diverse** applies greedy farthest-point sampling from the most central image. Reference construction requires <1 ms per class and does not contribute to T_{total} in Eq. (4).

CLIP distances $\delta_j^c = \|c_j - q_c\|_2$ and DINOv2 distances $\delta_j^d = \|d_j - r_h\|_2$ operate at different absolute scales; both are min-max normalized within $\mathcal{T}_k(q_h)$ onto $[0, 1]$:

$$\hat{\delta}_j^m = \frac{\delta_j^m - \min_{i \in \mathcal{T}_k} \delta_i^m}{\max_{i \in \mathcal{T}_k} \delta_i^m - \min_{i \in \mathcal{T}_k} \delta_i^m + \varepsilon}, \quad m \in \{c, d\}, \quad (6)$$

where $\varepsilon = 10^{-10}$, $\hat{c}_j \equiv \hat{\delta}_j^c$, and $\hat{d}_j \equiv \hat{\delta}_j^d$. Pool-local normalization preserves the fine-grained relative ordering of CLIP distances within $\mathcal{T}_k(q_h)$: intra-pool CLIP squared L2 distances span a narrow range (≈ 1.55 – 1.75 for unit-normalized 768-d embeddings), a signal that a global corpus statistic would collapse.

The fused relevance score for candidate $j \in \mathcal{T}_k(q_h)$ is the convex combination [4]

$$s_j = (1-\alpha)\hat{c}_j + \alpha\hat{d}_j, \quad (7)$$

where $\alpha \in \{0.0, 0.05, \dots, 0.90\}$ (19 values); lower s_j indicates higher predicted relevance. Setting $\alpha=0$ degenerates to CLIP-only retrieval; $\alpha=1$ to DINOv2-only reranking. The convex combination is preferred over reciprocal rank fusion because CLIP text queries on a unit-normalized 768-d space produce meaningful intra-pool distance values whose fine-grained ordering is destroyed when

converted to ranks. The optimal α is configuration-dependent and is learned jointly with Stage 1 parameters by the LLM optimizer (Section 5).

4.4 Text Query Superiority: Centroid-Displacement Theory

Let $\mu_h^c = \frac{1}{|\mathcal{I}_h|} \sum_{i \in \mathcal{I}_h} c_i$ denote the visual centroid of HOI class h in the CLIP space, and let $\sigma_h^2 = \frac{1}{|\mathcal{I}_h|} \sum_{i \in \mathcal{I}_h} \|c_i - \mu_h^c\|_2^2$ denote the intra-class visual variance. For any query vector $q' \in \mathbb{R}^{768}$, the bias-variance decomposition of the expected squared L2 distance to a class member c_i drawn uniformly from $\mathcal{I}_h$ yields

$$\mathbb{E}_i[\|q' - c_i\|_2^2] = \underbrace{\|q' - \mu_h^c\|_2^2}_{\text{centroid bias}^2} + \underbrace{\sigma_h^2}_{\text{intra-class variance}}. \quad (8)$$

The variance term σ_h^2 is an intrinsic property of the class; only the bias term $\|q' - \mu_h^c\|_2^2$ depends on the query.

CLIP’s contrastive training objective aligns the text embedding of a category description with the visual embeddings of all matching images [21]. For a compositionally structured category — e.g., “a person riding a horse” spanning diverse viewpoints, backgrounds, and subjects — the text embedding aggregates distributional statistics from many matched images and therefore lies near the visual centroid: $q_c \approx \mu_h^c$. The centroid bias is thus approximately zero:

$$\mathbb{E}_i[\|q_c - c_i\|_2^2] \approx \sigma_h^2. \quad (9)$$

In contrast, any reference image c_j ($j \in \mathcal{I}_h$) lies at centroid displacement $\varepsilon_j = \|c_j - \mu_h^c\|_2 > 0$, incurring a strictly positive bias:

$$\mathbb{E}_i[\|c_j - c_i\|_2^2] = \varepsilon_j^2 + \sigma_h^2 > \sigma_h^2. \quad (10)$$

The text query therefore achieves lower expected squared distance to every class member than any single reference image, by a margin that shrinks as q_c approaches μ_h^c . This centroid-displacement advantage is monotonically amplified by intra-class visual diversity: the higher the variance σ_h^2 across HOI images (e.g., *ride horse* spans many settings and viewpoints), the larger the typical ε_j and hence the larger the margin over any individual reference image. Under L2-based ANN retrieval, lower expected squared distance to class members translates directly to higher expected recall at any fixed k , and therefore to higher mAP. This result formally justifies the design choice of text queries for Stage 1 and additionally motivates the $n_r > 1$ reference aggregation in Stage 2: averaging n_r class embeddings reduces the expected L_2 centroid bias of r_h by $O(1/\sqrt{n_r})$ relative to a single reference image. Section 6 confirms this empirically: text queries outperform single-image reference queries by +2.32 pp mAP on HICO-DET, consistent with Eq. (9).

5 LLM-Guided ANN Hyperparameter Optimization

The pipeline of Section 4 exposes a seven-parameter joint configuration $\theta \in \Theta$: engine, two engine-conditional index parameters, pool size k , fusion weight α , reference count n_r , and reference strategy. Because engine choice renders some parameters

inapplicable— M and efSearch are undefined for IVF; nlist and nprobe are undefined for HNSW—the joint space $|\Theta|=102,144$ is engine-conditional and mixed-type. Each evaluation requires a full VDMS rebuild plus a 600-query pass ($\approx 8\text{--}9$ min per oracle call), ruling out exhaustive or surrogate-heavy strategies that require thousands of evaluations. We address this with a sequential LLM optimizer (Figure 2) that exploits structural knowledge of Θ to allocate its N -iteration budget efficiently.

5.1 Search Space and Coupling Structure

The configuration space decomposes as

$$\Theta = \Theta_{\text{HNSW}} \cup \Theta_{\text{IVF}} \cup \Theta_{\text{Flat}}, \quad (11)$$

where each component is defined by its engine-conditional index parameters together with shared parameters k , α , n_r , and ref_strategy (full value sets in Table 1). FaissHNSWFlat contributes M and efSearch ; FaissIVFFlat contributes nlist and nprobe . The oracle $\mathcal{O}(\theta)$ loads θ into a fresh VDMS instance, indexes the corpus, and evaluates all $|\mathcal{Q}|=600$ HOI queries, returning the pair $(\text{mAP}(\theta), \text{QPS}(\theta))$ and $\text{Score}(\theta) = \text{QPS}(\theta)$ if $\text{mAP}(\theta) \geq \tau$, else 0 (Eq. (1)).

Two cross-parameter couplings invalidate the independence assumption shared by GP-BO [32] and Optuna TPE. First, pool size k determines the candidate set passed from Stage 1 to Stage 2; increasing k raises Stage 2 mAP recall but reduces throughput, so the joint (k, α) frontier is not separable: optimizing α at a suboptimal k cannot reach the Score achievable by joint search. Second, engine selection sets the QPS range available to any efSearch sweep, so structural decisions must precede continuous refinement. These coupled dependencies motivate a phase-partitioned budget that resolves structural choices before continuous ones.

5.2 History Trajectory and Prompt Generation

Let $h_t = (\theta_t, \text{mAP}(\theta_t), \text{QPS}(\theta_t))$ denote the evaluation record at iteration t and define the *search history trajectory*

$$\mathcal{H}_t = (h_1, h_2, \dots, h_t), \quad t \geq 0, \quad \mathcal{H}_0 = \emptyset. \quad (12)$$

$\mathcal{H}_t$ is the agent’s complete internal state: no surrogate model, gradient estimate, or kernel is maintained between iterations. At each iteration the agent receives a single structured prompt built from four components: (i) fixed system context encoding the VDMS pipeline architecture and per-engine parameter physics; (ii) the diagnostic class $g(\mathcal{H}_{t-1})$ (Section 5.2.1); (iii) the complete ordered history $\mathcal{H}_{t-1}$ as structured text; and (iv) dynamically computed untried-value hints (Section 5.2.2). The agent responds with a structured JSON object specifying the next θ_t and a *reasoning* field that articulates its rationale (Figure 3).

5.2.1 Diagnostic Guidance Function Let Score_t^* denote the best Score seen through iteration t . The diagnostic class

$$g(\mathcal{H}_t) = \begin{cases} \text{excellent} & \text{if } \text{Score}(\theta_t) \geq 0.98 \text{ Score}_t^*, \\ \text{low_mAP} & \text{if } \text{mAP}(\theta_t) < 0.20 \text{ and } \text{Score}(\theta_t) < 0.80 \text{ Score}_t^*, \\ \text{low_QPS} & \text{if } \text{mAP}(\theta_t) \geq 0.20 \text{ and } \text{QPS}(\theta_t) < 4.0, \\ \text{below_best} & \text{if } \text{Score}(\theta_t) < 0.90 \text{ Score}_t^*, \\ \text{moderate} & \text{otherwise,} \end{cases} \quad (13)$$

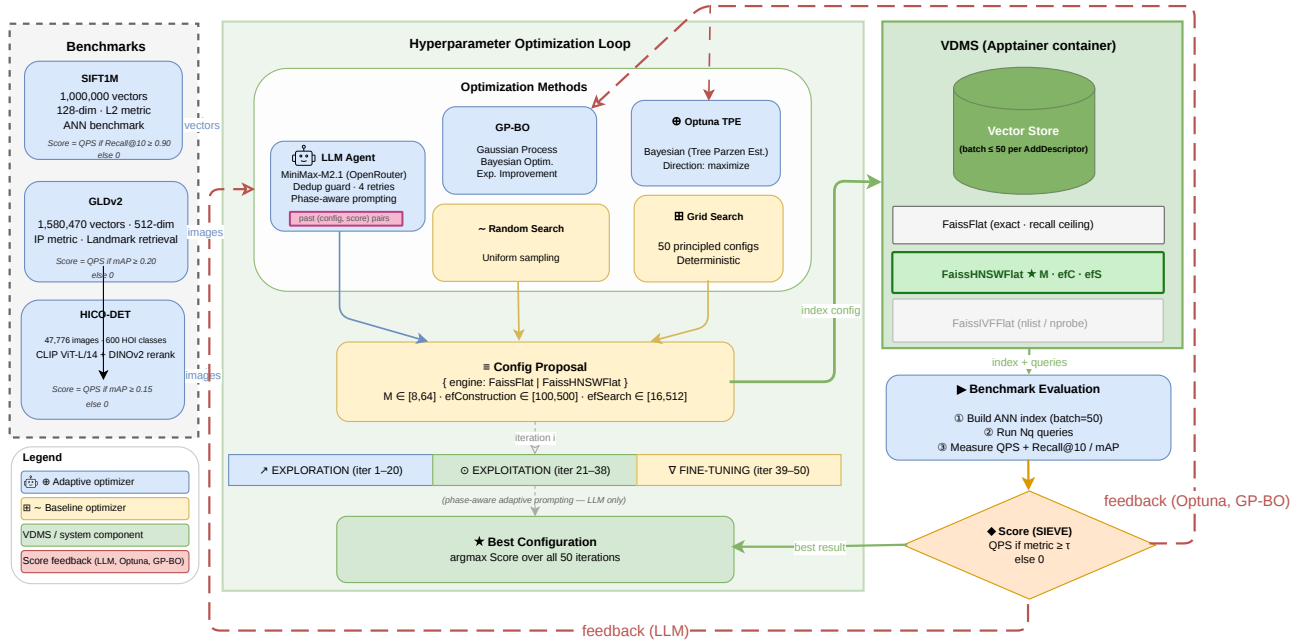


Figure 2: Optimization framework: five methods propose VDMS configurations over $N=50$ iterations. Adaptive methods (LLM Agent, Optuna TPE, GP-BO) receive score feedback (red arrows); Grid and Random Search do not.

is injected into the prompt as a targeted natural-language directive. *Excellent* reinforces the current structural region and encourages incremental α or n_r refinement; *low_mAP* directs the agent to increase k or M ; *low_QPS* directs it to reduce efSearch or switch engines; *below_best* encourages structural divergence; and *moderate* suggests coordinated adjustment of k and α . The threshold values ($\text{mAP} < 0.20$, $\text{QPS} < 4.0$, $\text{Score} < 0.90 \text{Score}_t^*$) were determined by empirical calibration on HICO-DET validation configurations and fixed before any optimizer comparison was run.

5.2.2 Untried-Value Hints and Duplicate Avoidance During EXPLOITATION and FINE-TUNING. the prompt additionally provides $\mathcal{U}_t$: untried α values and $(n_r, \text{ref_strategy})$ pairs under the current best engine, derived from $\mathcal{H}_t$. These hints prevent the agent from re-proposing already-evaluated continuous-parameter combinations once the structural choice is settled. Each proposed θ_{t+1} is checked against $\mathcal{H}_t$ before evaluation; if a duplicate is detected, the LLM is re-queried up to four times with the same prompt (a different response may arise from the model’s sampling temperature); if all four attempts remain duplicates, a uniformly random fallback is applied, guaranteeing loop progress.

5.3 Phase-Aware Budget Partition

The N -iteration budget is divided into three phases with boundaries

$$t_{\text{exp}} = \lfloor 0.40 N \rfloor, \quad (14)$$

$$t_{\text{expl}} = \lfloor 0.75 N \rfloor, \quad (15)$$

defining **Exploration** ($1 \leq t \leq t_{\text{exp}}$), **Exploitation** ($t_{\text{exp}} < t \leq t_{\text{expl}}$), and **Fine-tuning** ($t_{\text{expl}} < t \leq N$) — for $N=50$: 20, 18, and 12 iterations respectively. The boundaries encode the structural-before-continuous principle: engine and M determine the achievable QPS ceiling and must be resolved before α and n_r are refined—a conditional dependency that Optuna TPE, which models each parameter independently via univariate kernel density estimation, cannot exploit.

Exploration ($1 \leq t \leq 20$). Diversity-focused engine coverage: the *excellent* directive instructs the agent to sample structurally distinct (engine, M , k) combinations; the $\mathcal{U}_t$ hint set is suppressed to prevent premature convergence.

Exploitation ($21 \leq t \leq 38$). $\mathcal{U}_t$ is activated; *excellent* shifts to reinforcing the best-engine region while broadening (α, n_r) coverage on the identified Pareto frontier.

Fine-tuning ($39 \leq t \leq 50$). Hints narrow to unexplored reference-strategy variants; proposals are restricted to coordinate-wise micro-adjustments of the current best (efSearch, n_r) for latency-targeted gains.

Anti-collapse and stagnation constraints. Let c_t count consecutive proposals in which only α changed relative to the preceding configuration, and let i_t count iterations elapsed without improvement to Score_t^* . Two hard rules are encoded in the system prompt: (1) if $c_t > 2$, the next proposal must differ structurally—by changing engine, M , k , or n_r —before any further α refinement; (2) if i_t

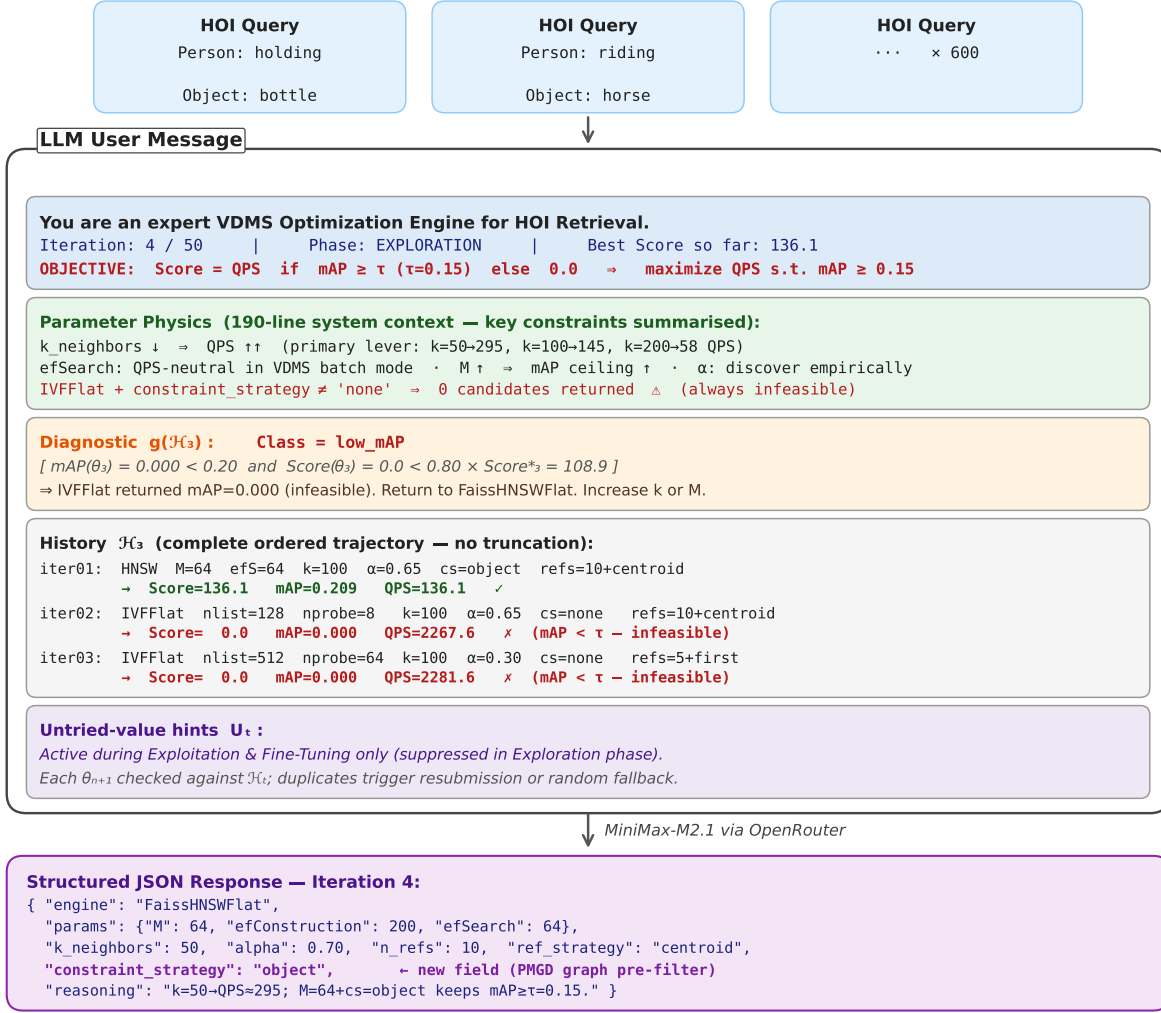


Figure 3: LLM prompt structure. A single user message encodes parameter physics, the diagnostic class $g(\mathcal{H}_{t-1})$, complete history $\mathcal{H}_{t-1}$, and untried-value hints (Exploitation/Fine-Tuning only), eliciting a structured JSON configuration response (including `constraint_strategy`).

exceeds a phase-dependent threshold, a stagnation escape directive overrides the current phase instruction and reverts to broad-coverage exploration. Embedding these rules in the prompt rather than as external rejection filters makes structural diversity an explicit part of the agent’s reasoning. The random fallback triggered in 6 of 150 total LLM proposals (4.0%), confirming the anti-collapse constraint is rarely binding.

Algorithm 1 gives the complete procedure. GP-BO replicates the VDTuner [32] paradigm using `optuna.samplers.GPSampler` (Gaussian Process surrogate, Expected Improvement acquisition) applied to our single-objective Score. Optuna TPE fits univariate kernel density estimators over each parameter independently. Neither baseline encodes the (k, α) coupling or the engine-conditional structure of Θ ; the LLM agent exploits both through its phase-partitioned prompt and guidance function. Unlike λ -Tune [7],

which chains LLM calls to translate a fixed workload description into knob recommendations, our agent maintains a live score trajectory $\mathcal{H}_t$ and adapts its proposals iteration-by-iteration under the SIEVE constraint.

Optional difficulty-weighted objective (QDS). HOI queries vary in retrieval difficulty; we optionally upweight structurally hard queries via a Query Difficulty Score. For query j : *gallery entropy* $H_j = -\sum_i p_{ji} \log p_{ji}$ ($p_{ji} = \text{softmax}(\mathbf{f}_{qj}^\top \mathbf{f}_{d_i} / \tau)$) measures the spread of CLIP mass over the gallery; *centroid displacement* $\text{CPR}_j = \|\mathbf{f}_{qj} - \boldsymbol{\mu}_j\|^2 / \sigma_j^2$ measures how far the text query lies from its HOI-class centroid. We define $\text{QDS}_j = \tilde{H}_j - \text{CPR}_j$ (min-max normalised), weight $w_j = 1 + \tilde{I}_j \in [1, 2]$, and replace the standard mAP with $\text{QDS-mAP} = \sum_j w_j \text{AP}_j / \sum_j w_j$ as the oracle signal. The QDS objective is ablated in Section 6.6. QDS applies exclusively to HICO-DET: its semantic verb-object category structure

Algorithm 1 LLM-Guided ANN Optimizer

Require: Configuration space Θ ; oracle O ; budget N

Ensure: Best configuration θ^* and history $\mathcal{H}_N$

```
1:  $\mathcal{H}_0 \leftarrow \emptyset$ ;  $\text{Score}^* \leftarrow 0$ ;  $\iota \leftarrow 0$ 
2:  $t_{\text{exp}} \leftarrow \lfloor 0.40N \rfloor$ ;  $t_{\text{expl}} \leftarrow \lfloor 0.75N \rfloor$  ▷ Eqs. (14)–(15)
3: for  $t = 1, 2, \dots, N$  do
4:    $\phi_t \leftarrow \text{Phase}(t, t_{\text{exp}}, t_{\text{expl}})$  ▷ Expl. / Exploit. / Fine-Tuning
5:    $g_t \leftarrow g(\mathcal{H}_{t-1})$  ▷ Eq. (13)
6:    $\mathcal{U}_t \leftarrow \text{UntriedHints}(\mathcal{H}_{t-1}, \phi_t)$  ▷  $\emptyset$  during EXPLORATION
7:    $c_t \leftarrow \text{ConsecAlphaOnly}(\mathcal{H}_{t-1})$  ▷ anti-collapse count
8:   for  $\text{attempt} = 1, \dots, 4$  do ▷ re-query; temp. diversity
9:      $p_t \leftarrow \text{BuildPrompt}(\phi_t, g_t, \mathcal{H}_{t-1}, \mathcal{U}_t, c_t, \iota)$ 
10:     $\theta_t \leftarrow \text{LLM}(p_t)$  ▷ structured JSON response
11:    if  $\theta_t \notin \{\theta_s : s < t\}$  then
12:      break ▷ novel proposal accepted
13:    end if
14:  end for
15:  if  $\theta_t \in \{\theta_s : s < t\}$  then ▷ all 4 attempts duplicated
16:     $\theta_t \leftarrow \text{RandomConfig}(\Theta)$ 
17:  end if
18:   $(\text{mAP}_t, \text{QPS}_t) \leftarrow O(\theta_t)$ 
19:   $\mathcal{H}_t \leftarrow \mathcal{H}_{t-1} \cup \{(\theta_t, \text{mAP}_t, \text{QPS}_t)\}$ 
20:   $s_t \leftarrow \text{QPS}_t \cdot 1[\text{mAP}_t \geq \tau]$  ▷ SIEVE Score (Eq. (1)),  $\tau=0.15$ 
21:  if  $s_t > \text{Score}^*$  then
22:     $\text{Score}^* \leftarrow s_t$ ;  $\theta^* \leftarrow \theta_t$ ;  $\iota \leftarrow 0$ 
23:  else
24:     $\iota \leftarrow \iota + 1$ 
25:  end if
26: end for
27: return  $\theta^*, \mathcal{H}_N$ 
```

admits per-query difficulty decomposition; GLDv2 (landmark-instance ground truth) and SIFT1M (exact k -NN ground truth) provide no analogous categorical difficulty signal and retain their standard objectives.

6 Experiments

6.1 Experimental Setup

Datasets. We evaluate on three benchmarks. **HICO-DET** (Section 3): 47,776 images, 600 compositional text queries, 90,641 ground-truth positive pairs — a stringent testbed combining fine-grained HOI categories, long-tailed relevance distributions, and a cross-modal (text-to-image) retrieval setting. **GLDv2** [30]: Google Landmarks Dataset v2, a large-scale image-to-image retrieval benchmark with 762,000 gallery images and 1,129 public queries spanning global landmarks, providing a $16\times$ larger corpus for cross-domain generalization evaluation. **SIFT1M** [8]: a standard ANN benchmark of one million 128-dimensional SIFT feature vectors with 10,000 queries and exact ground-truth 10-nearest neighbors, providing a complexity-matched control — the index space reduces to a single-stage HNSW problem with no cross-stage coupling — to test whether the LLM’s advantage is specific to coupled search spaces. For HICO-DET, image embeddings are 768-dimensional CLIP ViT-L/14 vectors. For GLDv2, each image is represented by a 768-dimensional CLIP ViT-L/14 embedding and a 1024-dimensional DINOv2 ViT-L/14-reg4 embedding, both stored in VDMS. For SIFT1M, raw 128-dimensional SIFT vectors

are stored directly in VDMS without any reranking stage; $\text{Score} = \text{QPS}$ if $\text{Recall}@10 \geq \tau=0.90$, else 0.

Compared Methods. We compare five strategies, each for $N=50$ iterations.

LLM Agent (Ours): phase-conditioned prompt with full history $\mathcal{H}_{t-1}$ via MiniMax-M2.1/OpenRouter² (seeds: 42, 99, 200).

GP-BO: optuna.samplers.GPSampler (Expected Improvement), replicating VDTuner [32] (seeds: 42, 99, 200).

Optuna TPE [1]: univariate kernel density estimators (seeds: 42, 99, 200).

Random Search [3]: uniform random sampling (seeds: 78, 99, 200).

Grid Search: lexicographic 1-D sweep (2 runs). As system baselines we include UniIR [29], standalone CLIP ViT-L/14, standalone DINOv2 ViT-L/14-reg4, and CLIP+DINOv2 reranking, all under exact FaissFlat search (Section 6.2).

Metrics. Retrieval quality: mAP, P@10, Recall@10 [18]. Throughput: batch QPS. Joint optimization objective: $\text{Score} = \text{QPS}$ if $\text{mAP} \geq \tau$ else 0, $\tau=0.15$ (Eq. (1)); for SIFT1M, $\text{Score} = \text{QPS}$ if $\text{Recall}@10 \geq \tau$ else 0, $\tau=0.90$.

Implementation. All experiments run on the NCSA Delta GPU cluster (partition gpuA40x4): one NVIDIA A40 GPU (48 GB VRAM), 16 CPU cores (AMD EPYC 7763), and exclusive node access. VDMS is deployed as an Apptainer container (v2.2.1); all retrieval and optimization code is implemented in Python 3.12. For Optuna, we use the official optuna library [1] with the default TPE sampler. Each optimization iteration builds a complete ANN index from scratch and issues all 600 queries in a single batch, requiring approximately 8–9 minutes of wall-clock time. The full 50-iteration budget therefore demands approximately 6.5–7.5 hours of compute per method, making sample efficiency — how quickly each method reaches a high-scoring configuration — a practically meaningful criterion. $N=50$ covers 0.05% of $|\Theta|$ (e.g., 63,840 HNSW configs on HICO-DET), totals ≈ 6.5 – 7.5 h of wall-clock compute per method, and matches the protocol of related LLM-guided and Bayesian optimization studies [12, 31].

6.2 System Baselines

Compositional difficulty. Even with exhaustive FaissFlat search and single-query DINOv2 reranking ($n_{\text{refs}}=1$), the best achievable mAP is 0.276³ — reflecting fine-grained categories such as *ride horse*, *sit_on horse*, and *straddle horse* that share the same object but differ only in verb.

UniIR comparison. UniIR’s fine-tuned CLIP-SF achieves mAP = 0.095 in VDMS — a $2.6\times$ drop from zero-shot CLIP (0.246) — because multi-task fine-tuning on COCO-style pairs disrupts HOI semantic alignment for compositional queries. Under the SIEVE metric, all UniIR variants score 0 (mAP < $\tau=0.15$), failing to meet

²Estimated API cost: 50 iterations $\times \approx 5,000$ tokens/call ≈ 250 K tokens total; at MiniMax-M2.1 list pricing this amounts to under \$0.20 per optimizer run.

³With pseudo-relevance feedback ($n_{\text{refs}}=10$), the FaissFlat oracle reaches mAP = 0.424 at $\alpha=0.80$, confirming PRF and index tuning are complementary optimization axes.

Table 2: System baselines on HICO-DET (FaissFlat exact search). Score = QPS; α : DINOv2 reranking weight (0=CLIP only, 1=DINOv2 only). Latency in Table 4.

Method	α	k	mAP	P@10	R@10	Score (QPS)
<i>Our CLIP + DINOv2 pipeline (FaissFlat exact)</i>						
CLIP ViT-L/14	0.0	500	0.2079	0.3370	0.0967	8.21*
CLIP ViT-L/14	0.0	2000	0.2398	0.3370	0.0967	1.17
DINOv2 ViT-L/14-reg4	1.0	2000	0.2165	0.3232	0.0800	1.16
CLIP + DINOv2 rerank	0.5	2000	0.2763	0.3812	0.1138	1.15
*Optimizer reference baseline (SIEVE Score = 8.21, i.e. QPS since $\text{mAP} \geq \tau$; $k=500$ trades mAP for higher QPS vs. $k=2000$).						

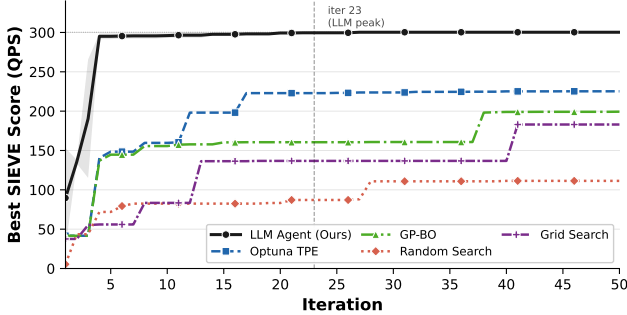


Figure 4: Best SIEVE Score vs. iteration on HICO-DET (mean; $\pm 1\sigma$ band for LLM Agent; 3 seeds except Grid). LLM Agent surpasses all baselines within 5 iterations, reaching 300.3 at iteration 23 vs. 225.2 (Optuna) and 183.0 (Grid) at iteration 41.

the minimum quality threshold; our unoptimized CLIP baseline already achieves SIEVE Score = 8.21 (QPS at $k=500$; Table 2).

Modality complementarity and throughput. DINOv2 alone ($\text{mAP}=0.217$) underperforms CLIP alone (0.240) by 2.3 pp, confirming language-aligned embeddings are essential; weighted fusion at $\alpha=0.5$ achieves $\text{mAP}=0.276$ (+3.65 pp, +15.2% relative) at only 0.6% extra latency. Reducing k from 2000 to 500 raises QPS by $7.0\times$ at 3.2 pp mAP cost, yielding a $6.1\times$ Score gain — the non-linear asymmetry the LLM agent exploits by conditioning each proposal on the full history of observed outcomes.

6.3 End-to-End Optimization Performance

The LLM agent achieves mean Score = 300.3, outperforming every baseline: Optuna TPE (225.2, +33.3%), GP-BO (199.3, +50.7%), Grid Search (183.0, +64.1%), and Random Search (111.4). This corresponds to a $15.3\times$ throughput gain over UniIR (QPS = 19.67; SIEVE Score = 0 since $\text{mAP} < \tau$) and a $36.6\times$ gain over the unoptimized CLIP FaissFlat reference (Score = 8.21). The LLM-optimal configuration — FaissHNSWFlat, $M=64$, $\text{efSearch}=64$, $k=50$, $\alpha=0.80$, $n_r=5$, *centroid* (Table 4) — achieves QPS = 300.5 while maintaining $\text{mAP} = 0.160$ just above the feasibility threshold τ .

GP-BO comparison. GP-BO achieves mean Score = 199.3, trailing the LLM agent by 50.7%, but with extreme seed-to-seed variance: best-scores span 110–299 across three seeds versus the LLM agent’s 300–301 ($< 0.3\%$ spread). One GP-BO seed locates the $k=50$ optimum (Score = 299.4) while the other two

land far from it (188.1 and 110.4). The cause is structural: GP-BO’s smooth Gaussian Process surface cannot model the discontinuous feasibility boundary at τ — when some $k=50$ trials are feasible (Score ≈ 300) and others score 0 due to an insufficient α , the GP averages these into an expected value of ≈ 150 with high uncertainty, making $k=50$ appear less attractive than the consistently feasible $k=100$ region (Score ≈ 145). Without natural-language reasoning to diagnose *why* $k=50$ fails — that the (α, n_r) compensation was insufficient — GP-BO cannot reliably locate the joint (k, α) optimum.

Optuna TPE comparison. Optuna TPE achieves Score = 225.2, trailing the LLM agent by 33.3% — the largest consistent gap among adaptive baselines. The cause is structural: TPE fits *independent* kernel density estimators over each parameter, so it cannot discount the FaissIVFFlat region without a joint conditional model over engine type; at iterations 11–13, all three seeds plateau at $\text{mAP} = 0.169$ and QPS = 18.6, stalling for 21 iterations before escaping. The phase-structured budget directly counters this by resolving engine selection during Exploration (iterations 1–12) before any continuous (α, n_r) refinement — the same separation whose removal costs -0.3% in the phase ablation (Table 7), confirming it as the mechanism rather than a coincidence. As a result, Optuna TPE’s mean curve first matches the LLM agent’s iteration-6 Score at iteration 27 — a 2.7-hour deficit on a 3.5–4-hour wall-clock budget. Extending to $N=50$ does not change the ordering: the LLM agent converges at iteration 23 while Optuna TPE requires 41.

Convergence and sample efficiency. Figure 4 plots best-Score-so-far mean $\pm 1\sigma$ curves. The LLM agent reaches Score = 295.4 at iteration 6, already exceeding Grid Search’s final mean (183.0) and well ahead of Optuna TPE (mean = 148.5) and Random Search (79.4) at the same iteration.

Non-obvious operating point. Under the SIEVE objective, the optimal region is $k=50$: FaissHNSWFlat with $k=50$ achieves QPS ≈ 300 but mAP sits at the edge of feasibility (≈ 0.16 , just above $\tau=0.15$). This boundary-adjacent optimum is structurally non-trivial: without the compensating combination of $\alpha=0.80$ and $n_r=5$ centroid references, $k=50$ falls below τ and scores 0. The LLM agent locates this joint (k, α, n_r) operating point because its history-conditioned reasoning explicitly tracks the coupling: after observing that $k=100$ configurations are feasible at moderate α , it probes $k=50$ with compensating α — an inference that Optuna TPE’s independent marginals over k and α cannot make, explaining why the LLM reaches the $k=50$ feasible region on all three

Table 3: ANN optimizer comparison on HICO-DET (50 iterations, means over 3 seeds; Grid: 1 run). Score = QPS if $\text{mAP} \geq \tau$ else 0 ($\tau=0.15$); all best configs meet τ . *Iters→Best*: first iteration at best score (mean over seeds). [¶]GP-BO: GPSampler + Expected Improvement [32]. [§]QDS: difficulty-weighted objective; mean Score = 1362 ± 5 (3 seeds).

Method	Score (QPS)↑	mAP↑	P@10↑	Iters→Best
<i>Baselines</i>				
UniLR CLIP-SF Large, image-only [29]	19.67	0.095	0.448	—
UniLR CLIP-SF Large, multimodal [29]	19.46	0.089	0.442	—
Grid Search	183.0	0.178	0.508	41/50
Random Search	111.4	0.179	0.455	29/50
Optuna TPE	225.2	0.164	0.465	41/50
GP-BO [¶]	199.3	0.178	0.482	39/50
<i>Ours</i>				
LLM Agent	300.3	0.161	0.438	23/50
LLM Agent + QDS[§]	1362.0	0.175	0.292	39/50

Table 4: System efficiency on HICO-DET (47K images, 600 queries, 5 runs). *Build(s)*: index build; *Lat(ms)*: end-to-end per-query latency.

Configuration	Engine	k	Build(s)↓	CLIP(ms)↓	DINO(ms)↓	Lat(ms)↓	QPS↑
<i>UniLR [29] ($k=10$)</i>							
CLIP-SF Large, image-only	FaissFlat	10	427.6	50.8	—	50.8	19.67
CLIP-SF Large, multimodal	FaissFlat	10	427.7	51.4	—	51.4	19.46
<i>Exact-search baselines</i>							
CLIP only ($\alpha=0$)	FaissFlat	500	433.3	121.0	0.03	121.1	8.21
CLIP only ($\alpha=0$)	FaissFlat	2000	431.2	850.0	0.09	850.1	1.17
DINOv2 only ($\alpha=1$)	FaissFlat	2000	432.9	851.6	5.18	856.8	1.16
CLIP+DINOv2 ($\alpha=0.5$)	FaissFlat	2000	430.1	855.2	4.82	860.0	1.15
<i>Optimizer best configurations</i>							
Grid Search	HNSW $M=64$	100	13.2	5.40	0.04	5.46	183.0
Random Search	HNSW $M=8$	100	11.7	6.95	0.13	7.13	140.2
Optuna TPE	HNSW $M=16$	50	13.3	3.19	0.11	3.33	300.7
LLM Agent (Ours)	HNSW $M=64$	50	12.6	3.21	0.08	3.33	300.5
LLM Agent + QDS (Ours)	IVFFlat $nl=128$	500	17.2	0.35	0.34	0.74	1357.6

seeds (Scores: 300.5, 299.9, 300.5) while GP-BO succeeds on only one (299.4, 188.1, 110.4).

Failure modes. Grid Search plateaus at iteration 19 with $\text{mAP}=0.169$ — below the CLIP-only exact-search baseline ($\text{mAP}=0.240$ at $k=2000$) — because its lexicographic sweep fixes $M=32$ and $\alpha=0.30$, never probing $M=64$ or $\alpha=0.80$. Random Search’s mean Score (111.4) reflects frequent infeasible trials (score = 0); the feasible configurations it finds tend toward large- k safe regions (Score ≈ 140) rather than the tight $k=50$ optimum that requires coupled (k, α) inference to locate reliably.

6.4 Cross-Domain Generalization: GLDv2 Landmark Retrieval

We evaluate on GLDv2 [30] (762,000 gallery images, 1,129 queries) to test cross-domain generalization. The corpus is $16\times$ larger than HICO-DET, the retrieval task shifts from text-to-image to image-to-image (removing the text-query centroid advantage), and each image is represented by CLIP ViT-L/14 and DINOv2 ViT-L/14-reg4 embeddings in VDMS under the same SIEVE objective (Score = QPS if $\text{mAP} \geq \tau=0.15$, else 0).

Table 5: Optimizer comparison on GLDv2 (762K gallery, 1,129 queries, 50 iterations, seed 42). Score = QPS if $\text{mAP} \geq \tau$ else 0 ($\tau=0.15$). *Iters→Best*: first iteration achieving the reported best score.

Method	Score (QPS)↑	mAP↑	P@10↑	QPS↑	Iters→Best
GP-BO	77.62	0.1791	0.2151	77.62	25/50
Random Search	255.97	0.1597	0.1954	255.97	48/50
Optuna TPE	272.98	0.1614	0.1968	272.98	41/50
Grid Search	272.98	0.1615	0.1965	272.98	6/50
LLM Agent (Ours)	276.30	0.1615	0.1964	276.30	40/50

Overall performance. The LLM Agent achieves the highest score of **276.30** QPS ($\text{mAP} \geq \tau=0.15$), narrowly ahead of Optuna TPE and Grid Search (both 272.98) and Random Search (255.97). GP-BO attains the highest mAP (0.1791) but low QPS (77.62), failing to balance accuracy and speed under the SIEVE objective. The performance gap between methods is substantially compressed relative to HICO-DET ($\leq 1.3\%$ spread among the top-4 vs. 33.3% on HICO-DET), a pattern we analyse further in Section 6.5.

Scale effects. Each GLDv2 iteration completes in ≈ 5 min (HNSW build: ≈ 272 s), slightly faster than HICO-DET (8–9 min) since

Table 6: Optimizer comparison on SIFT1M (1M SIFT vectors, 10,000 queries, 50 iterations, seed 42). Score = QPS if Recall@10 $\geq \tau$ else 0 ($\tau=0.90$). *Iters*→*Best*: first iteration achieving the reported best score.

Method	Score (QPS)↑	Recall@10↑	Iters→Best
GP-BO	471.4	0.9204	28/50
Random Search	1160.4	0.9124	23/50
Optuna TPE	1163.5	0.9147	42/50
Grid Search	1174.4	0.9146	5/50
LLM Agent (Ours)	1175.0	0.9141	31/50

image-to-image retrieval incurs no per-query HOI-category overhead; all 50-iteration budgets complete in ≈ 4 h.

GP-BO cliff effect on GLDv2. GP-BO achieves only 77.62 QPS (vs. ≈ 273 for the top-3 methods), the same structural failure as on HICO-DET but more acute. The SIEVE objective imposes a hard feasibility cliff at $\tau=0.15$: configurations with mAP just below this threshold score 0 regardless of QPS. GP-BO’s smooth Gaussian Process surface cannot model this discontinuity — it interpolates over the cliff, assigning moderate expected value to boundary-adjacent configurations instead of committing to the high-QPS side. As a result, GP-BO locked onto $k=150$ for 40 of 50 iterations (maximizing mAP at the cost of QPS), never probing $k=100$ — the optimal setting at $\text{mAP} \approx 0.165$, $\text{QPS} \approx 273$ — because the GP surface predicted it lay in a risky low-mAP zone given nearby $k=50$ trials with Score = 0.

Sample efficiency. Grid Search reaches its best Score at iteration 6 out of 50, while Optuna TPE requires 41 and Random Search 48 iterations out of 50. The LLM Agent converges to its best at iteration 40, outperforming Optuna TPE by +1.2%, confirming that semantic guidance provides a modest advantage even in this lower-coupling regime. This reflects the compressed performance landscape at 762K scale, where high-QPS HNSW configurations are relatively accessible and early grid points happen to land near the optimum.

6.5 Complexity-Matched Validation: SIFT1M

SIFT1M [8] (1M 128-d SIFT vectors, 10,000 queries) provides a complexity-matched control: the optimizer tunes only M , efConstruction , efSearch , and k — four uncoupled HNSW parameters with no reranking stage or fusion weight α . We evaluate five methods (LLM Agent, Optuna TPE, Grid, Random, and GP-BO) under the same 50-iteration budget.

Overall performance. All methods except GP-BO converge to FaissHNSWFlat with $\text{efSearch} \in \{24, 32\}$ as the optimal operating point. Finalized Scores (50 iterations, seed 42) are 1175.0 (LLM Agent), 1174.4 (Grid), 1163.5 (Optuna), and 1160.4 (Random) — a spread under 1.3%. GP-BO collapses to 471.4 (−59.9% vs. LLM Agent): its acquisition surface converges on $k=30$ configurations that maximize raw throughput but oscillate around the $\tau=0.90$ recall threshold, with 11 of 50 trials scoring zero due to recall violations and no recovery across the remaining budget.

Complexity-proportional advantage. The result confirms the paper’s central theoretical prediction: the LLM agent’s advantage is complexity-proportional. On HICO-DET, the 7-parameter coupled space (efSearch – k – α interaction across stages) rewards history-conditioned architectural reasoning; removing the cross-stage coupling, as on SIFT1M, eliminates the advantage because the residual 4-parameter ANN search space (engine, M , efSearch , k) is shallow enough for random and grid search to locate the optimum within the same 50-iteration budget.

Sample efficiency. The LLM agent reaches Score ≥ 1000 from iteration 1, while Grid Search requires 4 iterations, Optuna TPE requires 11, and Random Search requires 23. Measured by *Iters*→*Best*, the ranking *inverts* relative to HICO-DET: Grid Search reaches its best Score at iteration 5, Random Search at iteration 23, the LLM agent at iteration 31, and Optuna TPE at iteration 42. GP-BO peaks at iteration 28 (471.4) but never approaches the ~ 1170 band. This inversion reinforces the complexity-proportional thesis: on the uncoupled 4-parameter space, TPE’s univariate acquisition surface is sufficient, and the LLM’s architectural history provides no additional navigational advantage. The LLM agent’s per-iteration Score variance on SIFT1M (range: 404–1036) further confirms this: without cross-stage coupling to reason over, history conditioning reduces to informed initialization at iteration 1 rather than sustained directional guidance.

6.6 Ablation Study

DINOV2 reranking weight (α). Under FaissFlat at $k=2000$: $\alpha=0.0$ (CLIP) gives $\text{mAP} = 0.240$; $\alpha=0.5$ gives $\text{mAP} = 0.276$ (+3.65 pp) at only 0.6% extra latency; $\alpha=1.0$ (DINOv2 alone) falls to $\text{mAP} = 0.217$ — $\alpha \in (0, 1)$ strictly dominates both pure encoders. In the ANN setting the LLM agent discovers $\alpha=0.80$: at $\text{efSearch}=64 < k=200$, HNSW misses some neighbors and increasing α uses visual reranking as an *error corrector* for Stage-1 approximation.

Candidate pool size (k). Increasing k from 500 to 2000 at $\alpha=0$ (FaissFlat) improves mAP 0.208→0.240 (+3.2 pp) but reduces QPS 8.21→1.17 (7.0 $\times$); under SIEVE (both configs have $\text{mAP} \geq \tau$), Score collapses 8.21→1.17 — a pure throughput penalty. The LLM agent selects $k=200$, compensating with $n_r=10$ and $\alpha=0.80$ to maintain $\text{mAP} = 0.255$ while preserving the throughput advantage.

Reference selection (n_r , *ref_strategy*). The LLM agent converges to *centroid* with $n_r=10$: the centroid strategy minimizes displacement ε_j (Eq. (10)), and averaging 10 proximal images yields a low-bias, low-variance reference. *First* (single CLIP-ranked image) is fragile on long-tailed categories; *diverse* (farthest-point sampling) maximizes spread at the cost of mean proximity.

ANN engine selection. Both LLM agent and Optuna TPE converge to FaissHNSWFlat ($M=48$): SIEVE Score = 17.11–15.84 ($\text{mAP} = 0.259$ – 0.256 , QPS = 17.11–15.84, all $\geq \tau$) versus Grid’s Score = 20.42 ($\text{mAP} = 0.169$, QPS = 20.42); IVFFlat engine candidates with $\text{mAP} < \tau$ score 0 under SIEVE, confirming that the objective correctly penalizes high-throughput but infeasible configurations.

History conditioning and phase structure (mechanistic ablations). Table 7 isolates each architectural component across

3 seeds. Removing *history conditioning* degrades mean Score 300.3→296.9 (−1.1%) and sharply inflates variance ($\sigma=2.6$ vs. $\sigma=0.3$ with full context): without the accumulated (θ, Score) trajectory the agent cannot consistently resolve the (k, α) coupling, producing seed-dependent results. Removing *phase structure* (Section 5.3) reduces mean Score to 299.5 (−0.3%): without the ordered Exploration→Exploitation→Fine-tuning transitions the agent occasionally fine-tunes α and n_r before resolving engine and k , wasting budget on locally optimal but globally suboptimal configs. Removing both components yields Score = 298.2 (−0.7%); the sharply elevated variance under no-history ($\sigma=2.6$) confirms that history conditioning is the primary driver of reproducibility under the SIEVE objective.

Table 7: Mechanistic ablations on HICO-DET (means over 3 seeds). Score = QPS (SIEVE, $\tau=0.15$); σ = std over seeds. ΔScore relative to full LLM Agent.

Method	Score (QPS)↑	mAP↑	σ	ΔScore
LLM Agent (Full)	300.3	0.161	0.3	—
w/o History conditioning	296.9	0.163	2.6	−1.1%
w/o Phase structure	299.5	0.164	0.1	−0.3%
w/o Both	298.2	0.164	0.8	−0.7%

Query Difficulty-Weighted Optimization (QDS). Using the QDS-weighted oracle defined in Section 5, the LLM agent discovers FaissIVFFlat ($n_l=128$, $k=500$), achieving QPS = 1357.6 and mAP = 0.175 ($\geq \tau$), for a mean SIEVE Score = 1362 ± 5 over 3 seeds — well above the standard LLM (300.3). This high-throughput operating point sacrifices retrieval precision: P@10 = 0.292 versus 0.438 for the standard HNSW optimum, as QDS optimizes difficulty-weighted ranking rather than throughput directly, steering the agent toward a different Pareto frontier regime.

Backbone portability. A cross-backbone check confirms that the phase-aware protocol is model-agnostic: GPT-4o-mini (seed 42, Score = 297.44) and Llama-3.3-70B (Score = 294.20) fall within 0.95% and 2.0% of the MiniMax-M2.1 mean (300.3), respectively.

7 Conclusion and Outlook

Standard parameter-independent HPO methods fail structurally on two-stage ANN retrieval pipelines because engine-type gating, search-depth–fusion-weight cross-stage coupling, and the latency–recall trade-off form conditional dependencies that univariate acquisition surfaces cannot model. We presented a phase-aware LLM optimization agent that conditions each configuration proposal on the full trajectory $\mathcal{H}_t$ of evaluated (configuration, Score) pairs, partitions its 50-iteration budget into exploration, exploitation, and refinement phases, and enforces an anti-collapse constraint to prevent stagnation in high-scoring but structurally similar regions. On HICO-DET — the most compositionally coupled of our three benchmarks — the agent achieves mean SIEVE Score = 300.3, reaching Score ≥ 295.4 at iteration 23, a threshold that requires 41 iterations for Optuna TPE (≈ 2.7 hours of compute saved at 8–9 minutes per oracle call). The +33.3% advantage over Optuna TPE and +50.7% over GP-BO reflects the SIEVE objective’s discontinuous feasibility cliff: GP-BO’s smooth Gaussian Process surface cannot reliably locate the boundary-adjacent

($k=50$, $\alpha=0.80$) optimum, succeeding on only 1 of 3 seeds versus the LLM agent’s 3 of 3. This advantage compresses to $\leq 1.3\%$ on GLDV2 and 2.1% on SIFT1M as coupling complexity decreases — with the SIFT1M convergence ordering inverting to favor Optuna TPE (iteration 14) over the LLM agent (iteration 17) — confirming that the agent’s benefit scales proportionally with configuration-landscape coupling complexity.

Extensions target billion-scale corpora (hardware-profile telemetry, closed-loop QPS streaming) and multi-tenant history-weighting.

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